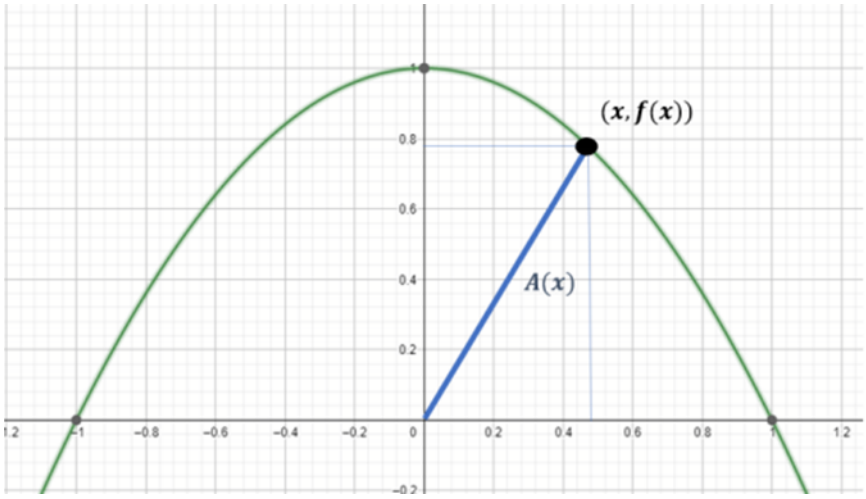
Lösung: Optimierung – Abstand zum Ursprung



a) Parabel: $f(x) = ax^2 + bx + c$

$P_1(-1,0) \Rightarrow f(-1) = a - b + c = 0$

$P_2(0,1) \Rightarrow f(0) = c = 1 \Rightarrow \underline{\underline{c = 1}}$

$P_3(1,0) \Rightarrow f(1) = a + b + c = 0$

$\Leftrightarrow \quad a - b = -1$

$a + b = -1 \quad +) \Rightarrow \underline{\underline{a = -1}} \quad \wedge \quad \underline{\underline{b = 0}}$

$\Leftrightarrow$ **Parabel: $f(x) = 1 - x^2$**

b) Abstand $A^2 = x^2 + f^2(x)$

$$A(x) = \sqrt{x^2 + f^2(x)} = \sqrt{x^2 + (1 - x^2)^2}$$

$$= \sqrt{x^2 + 1 - 2x^2 + x^4} = \sqrt{1 - x^2 + x^4}$$

$$A(x) = \sqrt{1 - x^2 + x^4} = (1 - x^2 + x^4)^{1/2}$$

$$A'(x) = \frac{1}{2}(1 - x^2 + x^4)^{-1/2} \cdot (4x^3 - 2x) = \frac{(4x^2 - 2)x}{\sqrt{1 - x^2 + x^4}}$$

Bei x_0 gilt: $A'(x_0) \stackrel{!}{=} 0$

$$\frac{(4x_0^2 - 2)x_0}{\sqrt{1 - x_0^2 + x_0^4}} = 0 \quad \Rightarrow \quad (4x_0^2 - 2)x_0 = 0$$

$$(x_0^2 - 1/2) \cdot x_0 = 0 \iff (x_0 - 1/\sqrt{2})(x_0 + 1/\sqrt{2}) \cdot x_0 = 0$$

Linearfaktorzerl. ↗

Nur $x_0 = +1/\sqrt{2} \in (0,1)$ liegt im Intervall.

$$x_0 \simeq 1/\sqrt{2} \approx 0{,}707$$

d)

$$A(x_0) = \sqrt{1 - x_0^2 + x_0^4} \approx \sqrt{1 - 1/2 + 1/4} = \sqrt{\frac{4 - 2 + 1}{4}}$$
$$= \sqrt{3/4} = \sqrt{3}/2$$

$\underline{\underline{A(x_0) \approx 0{,}87}}$ ← minimaler Abstand